

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
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Originating Company	Epic Atlantic Limited
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Specification for Piping

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Rev No.	Date	Description of Revision
R01	23-05-22	First Issue for Clients review
A01	20-12-22	Issued for Approval.

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	EA-BFPCL-GPMC-GGPL-FEED- MPP-SPC-015

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Material Class Specification
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-010	Specification for Pipes & Fittings
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-013	Specification for Flanges
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-012	Specification for Bolts and Nuts
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-014	Specification for Gaskets
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003	Specification for Piping Valves
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-011	Specification for Pipe Support.

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Abbreviations

Abbreviation	Meaning
AED	Anti-Explosive Decompression
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
API	American Petroleum Institute
BS	British Standard
CS	Carbon Steel
DBB	Double Block and Bleed
ENP	Electroless Nickel Plated
HFE	Human Factor Engineering
ISO	International Organisation for Standardization.
MESC	Material & Equipment Standard and Code
MMscf/d	Million standard cubic feet per day
MSS	Manufacturers Standardisation Society
MTO	Material Take Off
NDE	Non-Destructive Examination
NPT	Nominal Pipe Threaded
NPS	Nominal Pipe Size
RTJ	Ring Type Joint
SPE	Society of Petroleum Engineers.
SS	Stainless Steel
SW	Socket Weld

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

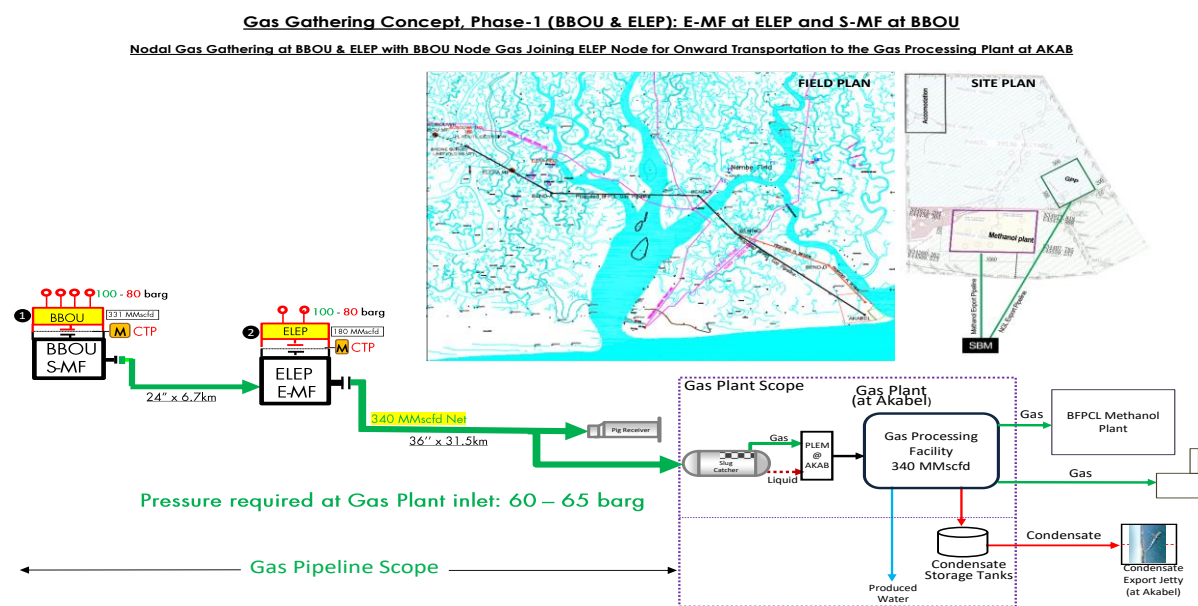


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This document presents the Piping Design specifications for the piping scope of BFPCL Gas Gathering Pipelines & Associated Infrastructure Phase 1. These design specifications ensure uniformity and consistency of the design by describing the following:

- Piping materials specification
- Piping Items design specification
- Piping layout requirements

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 31.38.01.25-Gen.	Piping – Process Design Requirements
DEP 31.38.01.11-Gen	Piping - General Requirements
DEP 31.38.01.24-Gen.	Piping – Engineering And Layout Requirements
DEP 31.38.01.31-Gen	Shop And Field Fabrication Of Piping
DEP 02.00.00.10-Gen.	Preparation Of Technical Drawings
MESC SPE 77/302	Valves – General requirements
DEP 74.00.10.10-Gen.	Shop And Field Pressure Testing Of Piping And Process Systems
DEP 31.38.01.29-Gen.	Pipe Supports.
DEP 30.00.60.20-Gen	Human Factors Engineering – Physical Workspace Design.
DEP.31.38.01.84-Gen	Piping Material Selection Index
ASME B1.1	Unified Inch Screw Threads (UN and UNR Thread Form)
ASME B1.20.1	Pipe Threads, General Purpose
ASME B16.5	Pipe Flanges and Flanged Fittings
ASME B16.9	Factory Made Wrought Steel Butt Welding Fittings
ASME B16.10	Face to Face and End to End Dimensions of Valves
ASME B16.11	Forged Steel Fittings, Socket Welding and Threaded
ASME B16.20	Metallic Gaskets for Pipe Flanges, Ring Joint, Spiral Wound and Jacketed
ASME B16.21	Non-Metallic Flat Gaskets for Pipe Flanges
ASME B16.25	Butt Welding Ends
ASME B16.34	Valves - Flanged, Threaded, and Welding End
ASME B16.36	Orifice Flanges
ASME B16.47	Large Diameter Steel Flanges NPS 26 through NPS 60

ASME B16.48	Line Blanks
ASME B18.2.1	Square and Hex Bolts and Screws (Inch Series)
ASME B18.2.2	Square and hexagonal nuts
ASME B31.3	Process Piping
ASME B36.10M	Welded and Seamless Wrought Steel Pipe
ASTM A105	Forgings, Carbon Steel for Piping Components
ASTM A106	Seamless Carbon Steel Pipe for High Temperature Service
ASTM A182	Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service
ASTM A193	Alloy-Steel and Stainless-Steel Bolting Materials for High Temperature
ASTM A194	Standard Specification for Carbon and Alloy Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both
ASTM A234	Pipe Fittings for Wrought Carbon Steel and Alloy Steel for Moderate and Elevated Temperatures
ASTM A370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
API 6D	Specification for Pipeline Valves
API 6FA	Fire Test for Valves
API 598	Valve Inspection and Test
API 599	Metal Plug Valves: Flanged & Welded Ends
API 600	Steel Gate Valves – Flanged and Butt-Welding Ends
API 602	Compact Carbon Steel Gate Valves
API 603	Class 150, Cast, Corrosion Resistant, Flanged End Gate Valves
API 607	Fire Test for Soft Seated Quarter-turn Valve
API 608	Metal Ball Valves: Flange, Threaded & Welding End
MSS SP-44	Steel Pipeline Flange
MSS SP-83	Class 3000# Steel Pipe Unions Socket Welding and Threaded
MSS SP-75	Specification for High-Test, Wrought, Butt-Welding Fittings
MSS SP-97	Integrally Reinforced Forged Branch Outlet Fittings-Socket Welding, Threaded, and Butt-Welding Ends

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract

- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Piping Design Requirements

3.1. General considerations

- Design life of the piping system in operation is 30 years.
- Piping systems shall be designed as per ASME B31.3 and ISO 15649. The DEP 31.38.01.11-Gen Piping - General Requirements.
- New piping class shall be developed for the pipeline in plot piping, valves, fittings, gaskets, bolts and nuts etc. around the pig trap areas that is not accommodated in DEP.31.38.01.84-Gen (Piping Material Selection Index). The development will be based on modifications to an existing piping class.
- Screw fittings shall be avoided as much as possible.
- Piping shall be routed so that the shortest practical length and a minimum number of fittings, branches and valves are used, consistent with a safe, operable, and maintainable layout.
- Allowance shall be made for any defined future requirement.
- Materials of piping components shall be in accordance with the Piping Material Classes.
- Piping items not included in the Piping Class shall be identified as special piping item

3.2. Selected Nominal Sizes

3.3. For all types of piping components the following limitations are imposed:

- DN 20 (NPS $\frac{3}{4}$) should not be used except for drain outlet.
- If pipes are to be supported on pipe ways, nominal pipe size on pipe ways shall not be less than DN 50 (NPS 2);
- Nominal pipe sizes 3½", 5", 7", 9" and 22" shall not be used. When non-standard size connection is inevitable, a transition to a standard pipe size shall be made as close to the connecting end as possible.
- Minimum line sizes to be adopted are as follows:

Line Type	Minimum Line Size
Drains/vents with no connections	$\frac{3}{4}$ "
Process lines	2"
Utility lines	1 $\frac{1}{2}$ "
Nozzles (e.g. for Pig Trap)	2"
Sample points	1"

3.4. Piping Layout Considerations

3.4.1. Generalities

- Piping shall be routed, such that the optimum piping layout is achieved in terms of process

requirements, ergonomics, operation, inspection and maintenance. Having considered these factors, the number of flanges, fittings, valves and welds shall be minimised.

- Except for identified utility systems (if any), all pressurised hydrocarbon piping shall be routed above ground.
- The routing shall take into account the movement of the piping and the possible movement of the equipment and/or structures to which the piping is connected or by which it is supported.
- Piping shall run in perpendicular directions. Changes in directions and elevations shall normally be at 90 degrees angles and no pockets shall be created where there are no drain arrangement. Diagonal and skewed angles of piping layout in horizontal plane should be avoided.
- Flanges in piping systems shall be installed only to facilitate maintenance and inspection.
- Spectacle blind and spades shall be located such that they can be installed or removed from the ground level or from platforms or walkway.
- Piping shall not touch other piping, structures, electrical cables or conduits, or instruments.
- Piping shall be routed such that overhead clearance (headroom) or stumbling hazards such as obstruction to walkways and other paths of travel or equipment movement are avoided.
- Where a piping system is connected to another system or to equipment of a higher design rating, the higher rating shall prevail for all piping components up to and including, the first block valve in the system of the lower rating, as indicated in the P&IDs.
- If a system operating above 0 °C (32 °F) is connected to a system operating at 0 °C (32 °F) or below, the first 1m (3.3 ft) of the higher temperature system shall be designed to the specifications of the lower temperature system.
- Valves shall be designed / located for ease of operation, installation, and maintenance

3.5. Clearance and Access

Workspace Design (As per DEP 30.00.60.20-Gen.)

<u>Minimum vertical clearances for unhindered travel:</u>	
By mobile crane	6m
By truck	4m
By lift truck	2.7m
On main road	6m.
On roads inside the unit	3.75m

Head clearance above walkway and platform	2.2.m
Minimum vertical space for operator standing posture	2.2.m
Below piping	2.2.m
Mobile lifting equipment in operation area	2.4m
Minimum Bottom of pipe elevation for pipes on pipe way	0.3m
Minimum vertical clearance between low points drain and grade	0.23m
Maximum height of single run ladder without intermediate rest platform	12m
Maximum run of single ladder without fall protection	3.m
Minimum access width for walkway on elevated work platform (distance behind the ladders up to the guardrail)	0.8.m
Minimum clearance in front of ladder	0.8m
<u>Horizontal clearances for unhindered travel:</u>	
For mobile maintenance equipment	3m
Between the shoulder of a road and the vertical projection of an item of equipment, or the end of a platform, structure, or pipe	0.5m
Minimum width to allow a man to pass freely	0.9m
Minimum width with Operating aisles	2.1m
Minimum width of platforms for the operation of a valve or instrument	0.610m
<u>Distance between Pipes and clearances</u>	
Between two pipes, or between a pipe and a flange	0.075m.
Minimum space for Bolt removal between flange and any obstruction	length of bolt + 0.025m

When necessary, the distance between pipes shall be increased to allow free movements of the

pipes.

3.6. Small Bore Piping

Since small bore branches ($\leq$ DN 40) to large bore piping are relatively susceptible to failure, the following points shall be incorporated in the piping design:

- Minimise the number of small-bore branches to piping.
- Branches shall not be located in removable spools unless it is impractical to do otherwise.
- Branches shall not be located in high stress areas.
- Branches should not be less than DN 50 (2")

3.7. Branch Fittings

A branch fitting connects a branch pipe to the run pipe.

Branches should be connected at 90 degrees to the run pipe in accordance with the General Piping Material Specification – Piping Class.

3.8. Vents and Drains

All pressurized hydrocarbon piping systems, equipment (e.g. pig trap) must be connected to a closed drain system.

Drain and vent connections for pressure testing shall be installed, without valves, at all low and high points in each piping system to facilitate venting and draining after pressure testing after construction or repair of the system

Hydrostatic vents and drains shall be installed in all piping and fitted with plugs according to the table of sizes below

Run size	Branch size
DN 20 through DN 50 (NPS $\frac{3}{4}$ through NPS 2)	DN 20 (NPS $\frac{3}{4}$)
DN 80 through DN 300 (NPS 3 through NPS 12)	DN 25 (NPS 1)

Process vents and drains shall be shown on the P&IDs. The number of vent and drain connections shall be minimized as far as possible.

All vent points shall be closed with a blind flange or plug if not used otherwise for process venting or draining

3.9. On-Line Probing, Coupon

Non-intrusive method shall be used for line inspection as against design based on fully retrievable probes. Clamp-On or strap-on Ultrasonic Intelligent Sensors shall be used for pig detection.

3.10. Block Valves

Ball valves for ON/OFF purposes installed shall be equipped with Gas Over Oil Operated Actuator (if considered).

3.10.1. Ergonomic Aspects of Valves

See appendix A

- The minimum distance between hand wheels and any obstruction shall be 75 mm.
- The location of valve hand-wheels and/or stems shall not obstruct walkways or platforms.
- Valves in horizontal pipes shall be positioned with their stem on or above the horizontal.
- Valves shall be permanently accessible for operation or maintenance.
- Valves which will be operated frequently to meet operating requirements shall be easily accessible from the ground, from a platform or from a fixed ladder.
- The horizontal centrelines of the hand wheels of valves operated from a platform must not be above a height of 1.5. meter.
- All valves 12" and above installed in horizontal positions shall be provided by the Vendor with support.
- All valves 10" and above or with a mass of 250 kg and greater shall be fitted with lifting lugs.
- The following manual Ball valves shall be fitted with gear operator as per DEP 30.00.60.20-Gen

ASME Class	Valve Type Ball.	
150	(NPS 10) and larger	(NPS 6) and larger
300	(NPS 8) and larger	(NPS 4) and larger
600	(NPS 6) and larger	(NPS 4) and larger
900	(NPS 4) and larger	(NPS 3) and larger

3.10.2. Relief Valves

Relief valve assemblies shall be installed in the vertical position, as close to the pressure source as possible and shall be provided with permanent platform access where required.

Relief valve piping shall be designed to withstand reaction forces and moments exerted when the by the valve is discharging.

Discharge piping from relief valves shall be self-draining away from the relief valve outlet and not be pocketed where there is the likelihood of liquids accumulating in the line.

Pressure-relief system shall be in accordance with DEP 80.45.10.10-Gen.

3.10.3. Control Valves and Manifold

All control valves shall preferably be installed in horizontal lines with the actuator in the vertical position. They shall be located as close as possible to the item of plant under control and shall be easily accessible from grade or permanent platform.

Where control valves are less than line size, reducing spools shall be made long enough to permit bolt removal. Consideration shall be made for the removal or withdrawal of valves or part of valves for maintenance.

Control valves shall be provided with block valves and a bypass valve that is capable of throttling. The block valves at each side of the control valve shall be of the straight-through type. The bypass line shall not be located directly above the control valve

3.11. Piping and Equipment Isolation

Where no specific tightness requirements are justified a single valve isolation shall be the standard way of separating systems with a bleed to verify that the isolation valve has seated and tightness has been reached before spading, and to provide a means of draining or depressurizing the volume between the spade/blind and isolation valve.

Double valve isolation shall be used in lines operating in ASME class rating 900# and above, regardless of the service. DBB arrangement shall be an assembly of individual valves with a bleed valve in between.

Spading ONLY shall be used to achieve positive isolation when there is intent for inspection and maintenance by personnel.

3.12. Piping Flexibility

Piping system shall be routed, supported, anchored, or guided so that thermal expansion/contraction, vibration or movements coming from any reason will not result in stresses in the piping or in the connected equipment in excess of those permitted by ASME B31.3 and the equipment design code, in order to prevent:

- Failure of piping component due to overstress
- Leakage at joints
- Excessive loads and movement on the concerned equipment, anchor points, flanged connections etc.

Piping shall be routed to take advantage of available natural flexibility. Depending of the recommendations of the stress analysis, flexibility loops shall be provided where natural flexibility in the piping system cannot be achieved.

3.12.1. Flexibility Analysis.

A formal comprehensive analysis of adequate flexibility shall be conducted for all piping system not meeting the criteria for exemption according to ASME B31.3 section 319.4. Generally, the flexibility of piping shall be assured by adequate routing (loops or change of direction).

3.13. Stress Analysis

Stress analysis will be carried out in accordance with ASME B31.3

As per DEP 31.38.01.11-Gen.section 3.5; Critical systems shall be identified to allow full stress analysis to be carried out, which will determine additional flexibility and position of critical supports (anchors, guides, etc.). Formal computer analysis shall be made using EMPLOYER approved software. If required in this project that the entire piping system be stress analysed, otherwise;

- Pipes DN 500 and larger
- in cases where differential settlement of equipment and/or supports is expected
- pipes subject to slug flow or water hammer
- pipes for two-phase flow
- flare pipes DN 100 and larger
- pipes subjected to relief loads

Refer to Critical line list for BBOU, AKAB and ELAP for list of process lines that shall be stress analysed.

Piping flexibility and stress analysis report shall be presented in EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-002, – 003 and 004 for BBOU, ELEM and AKAB respectively.

3.14. Pipe Support

Pipe support shall be provided to handle the effect of weight and forces from piping (pipes, valves etc.) introduced by service pressures and temperatures, vibration and expansion/contraction which could result in stresses in excess of those permitted by ASME B31.3.

The Selection of support type, design of the supporting elements, location of pipe support and installation shall be in accordance with the technical Specification for Pipe Support EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-011, DEP 31.38.01.29-Gen and DEP 31.38.01.11-Gen

Furthermore, the result of a computerised stress analysis showing points of high stresses, resultant load distribution and possible strain where supporting is required shall be used to affirm the final locations of all support.

As a minimum, all piping should be located on pipe-rack and concrete sleepers depending on the grade level of the installation.

Pipe support span for straight steel pipes and for bent steel pipe shall be in accordance with

Appendix B.

Support of valve 16" and above to be preferably installed just underneath the valve and to be "rest type" only.

Pipe supports shall be indicated on the piping General Arrangement Drawings and on Isometrics Drawings with identification numbers.

Pipe supports shall not block access for operation and maintenance.

Support details and drawing shall contain dimensioned sketches of the applicable standard pipe supports and bill of materials in accordance with DEP 31.38.01.29-Gen.

3.15. Hydraulic Bolt Tensioning

Hydraulic bolt tensioning shall be used for tightening the bolts of flanged connections of equipment and piping where operating conditions require the stress distribution in the bolts to be properly controlled to obtain a reliable joint.

Hydraulic bolt tensioning shall apply for all services in the following cases as per DEP 31.38.01.31-Gen Table 3.4:

Service	ASME Rating Classes	Bolt diameter
All	All	≥ 50.8mm

Bolting for use with bolt tensioners shall have the normal length increased by one bolt diameter. Bolt extensions beyond the nut shall be protected from corrosion by threaded metal caps

3.16. Platform Requirements

A permanent platform or standing surface shall be provided for all operational tasks not within reach or view but requiring the use of both hands and unobstructed view.

The layout and arrangement of platforms and others means of access must be such that the supervisory rounds can be made at various levels without requiring excessive climbing up or down.

Platforms over 8 m long shall be served by two separate accesses (stairs or ladders). These means of access shall be placed in such way that no point between two accesses is more than 20 m from, at least one of them and there shall be not blind lengths of platform exceeding 8 m.

3.17. Instrumentation

Piping with instrument connections shall be routed such that safe access to these connections is ensured; if necessary, platforms or walkways shall be provided.

All instrument devices shall be permanently accessible, located not more than 1.5m vertically above grade, platform or walkway.

In-line instrumentation shall be mounted in accordance with DEP 32.31.00.32-Gen.

Field-mounted instruments shall be installed considering the following aspects:

On-line instrument are to be mounted on or in the direct vicinity of the instrument connection,

Limited accessibility is acceptable for indicating instrument, providing that they can be properly read from a permanent accessible location,

Instrument shall not be subjected to excessive vibration or to mechanical stresses.

Instrument process connections on horizontal process lines shall be located at the top (vertically up or pointing upwards at an angle of up to 45 degrees from the vertical axis) to limit the blocking risk by solid, dirt or pipe scale.

Instrumentation connection on either equipment or piping requiring either single or double block and bleed facilities shall be made of an integral manifold. The side of the integrated manifold, with connection to the equipment or piping must be flanged whilst the instrument connection side may be threaded as required

The length of the branch should be minimised (fitting-to-fitting)

3.17.1. Instrument Access Requirements

Maintenance access is required to all instruments where access is required for instrument operation, the minimum access shall be provided by the following means, in order of preference:

Legend: A = Grade or Platform

B = Permanent Ladder

C = Portable Ladder

Type of Instrument	Access for Operation Required	Access Requirements	Remarks
Temperature Elements	No	A, B, C	
Test thermowells	Yes	A	Locate so thermometer can be readily installed.
Pressure gauges	No	A, B	Dial to be visible to operator
Temperature transmitter and switches (indicating)	Yes	A, B, C	Dial to be visible to operator

Type of Instrument	Access for Operation Required	Access Requirements	Remarks
Temperature transmitter and switches (blind)	Yes	A, B, C	
Press switches and transmitters (indicating)	Yes	A, B	Dial to be visible to operator
Press switches and transmitters (blind)	Yes	A, B	
Recorders and controllers located on stands	Yes	A, B	
other final control elements	Yes	A, B	
DP level and flow transmitters	Yes	A, B	
Level controller/transmitter (ext. displacer type)	Yes	A	
Level switch (ext. displacer type)	Yes	A	

3.18. Handling Considerations

The maximum weight to be manually lifted by one person shall be 23 kg, lifts requiring more than one person are subject to a material handling study. Handling philosophy shall be developed during detailed Engineering.

Valves and other heavy items shall be handled using a mobile crane or monorail. Valves shall be positioned ergonomically for easy operation and maintenance. If not accessible from ground level, an access platform shall be provided.

A special trolley shall be used for to/fro movement of the pig. A monorail gantry and pulley suspended chain hoist assembly shall be installed for the lifting and handling of the pig during launching operation.

3.19. Human Factor Engineering

Human factor engineering (HFE) is an important step in the initial design to eliminate "human error" resulting from mal-operation, fatigue, confusion, discomfort, false interpretation, bad communication, accessibility, etc. HFE considerations shall comply with the requirements of DEP 30.00.60.20-Gen.

The following are the major considerations for “Human Factor” in equipment layout and piping design

- Good layout and safe accessibility and egress from facilities, equipment, instruments, etc. in the process, utility and other areas of the platform.
- Correct labelling of equipment, instruments and critical valves, including its location from easy visibility point of view. Quick identification particularly during start-up, commissioning and emergency operations is very essential and important.
- All pressure, temperature, flow, and level indicators shall be so arranged as to be clearly readable from access decks or platform.
- Design of installation shall take into account the tasks to be performed by maintenance operators and where required by construction personnel, e.g. safety/relief valves/ instruments removals, actuators removal, filling operations, operations of valves, swinging of man-hole davits, etc. This includes provision of access, lifting equipment, tools, etc.
- Escape routes, muster area, safety equipment, hazards, etc. shall be clearly marked and identified.
- Ventilation provision for large vessels where human entry is involved.

3.20. Weld Spacing

The minimum spacing between welds is as follows:

Pipe with wall thickness up to 20 mm shall have a minimum 50 mm or 5 times wall thickness between welds, whichever is the greater.

Pipe with wall thickness greater than 20 mm shall have a minimum 100 mm between welds.

Branch connections, pressure and non-pressure attachments shall have a minimum of 50 mm or 5 times wall thickness between welds.

3.21. Specification and Pressure Class Breaks

Piping material specification breaks and pressure class breaks between two pressure ratings shall only take place downstream of isolation valves (or across a relief valve) and the valve shall be rated at the higher of the two pressures unless otherwise defined on the P&IDs.

In the case of adjacent valves, e.g. double block and bleed, all valves shall be rated at the higher pressure.

3.22. Pressure Testing

3.22.1. General

Pressure test safety considerations, preparations, procedures, test fluid quality, pressure test planning, and post testing activities shall be in accordance with the requirements of DEP 74.00.10.10-Gen..

All non-destructive testing and stress relieving shall be completed with acceptable results before any pressure testing is done.

Appurtenances with a maximum test pressure less than the piping system test pressure shall be either removed or positively isolated before pressure testing the piping system.

3.22.2. Hydrostatic Testing

Hydrostatic leak test shall be conducted for all piping system to ensure tightness at a pressure not less than 1.5 of design pressure in accordance with paragraph 345.4 of ASME B31.3..

Test fluid shall be a non-toxic and non-flammable liquid (usually water).

Carbon steel piping system should be hydrostatically tested with water containing < 250 mg/L (< 250 ppmw) chlorides.

The use of liquid other than water shall be subject to a chemical evaluation of liquid to determine its corrosion effect on the piping system and subject to approval of the EMPLOYERs corrosion engineer.

3.22.3. Pneumatic Testing

Pneumatic testing is not required and shall not be considered in accordance with DEP 74.00.10.10-Gen.

4. Piping Components

4.1. General

Selected materials in accordance with the Piping Classes shall be used wherever possible to avoid unnecessary effort during engineering, purchasing, construction, shipping and storing of piping components.

Piping components, fittings, valves, gaskets and bolting shall comply with the technical requirements of the component description plus the supplementary requirements. The use of special materials and/or piping components (including Special Piping Items) not included in the piping classes shall be minimized, and shall require the approval of the responsible piping material engineer.

If a required item is not covered by the piping classes and requires individual calculation or selection, an SP number shall be allocated and used on all relevant documents, including the P&IDs. Such items might include valve locks, non-standard valves, strainers, hoses, transition pieces, quick release coupling. etc.

4.2. Pipes and Fittings

Pipes and fittings shall comply with the project specification.

Reducing elbows, straight crosses, reducing outlet crosses and short radius 45° or 90° elbows shall be considered as special fittings. Such special items shall be used only to suit process or piping requirements if no other design is feasible.

Target Tees and blinded Tee connections shall be used for change of direction or in place of elbows where the effluent of the piping system contains substantial amount of sand or particles.

Note that short radius elbows have reduced allowable pressure ratings and require calculations to be made before utilization; refer to ASME B16.28.

All branch connections shall be in accordance with the branch connection table shown in the Piping Material Classes.

The use of concentric reducers shall be the general rule. An exception is the design for pig traps using eccentric reducers which is pipeline scope in this project. Refer to Pipeline Equipment Specification - Pig Launchers & Receivers EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-005 for details.

4.3. Flanges, Spectacle Blinds, Spades and Spacers

Flanges shall be in accordance with EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-013 Specification for Flanges and as described in the Piping Classes. Slip-on flanges may be used in accordance with the piping material classes for 150# for utility services

Bolted joints developed for special applications shall at least meet the requirements and service

rating of the lowest rated component in the relevant piping system, as defined in the piping line list.

Spectacle blinds and spacers shall be of the same material as the line class pipe. Spades for short duration and/or intermittent use shall normally be of carbon steel. Spectacle blinds, spades (i.e. ring blinds) and spacers shall be in accordance with ASME B16.48

4.4. Gaskets

Gaskets shall be as per EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-014 Technical Specification for Gaskets and the applicable piping classes.

ASME Class 150# and 300# Flat gaskets or spiral wound are generally required.

ASME Class 600# Spiral wound gaskets are generally required.

ASME 900# and Above Ring Joint, octagonal type.

Gasket materials and the use of centering rings and inner rings shall be in accordance with the Piping Classes.

Compressed asbestos fibre (CAF) gaskets shall not be used.

4.5. Valves

Valves shall comply with the Specification for Piping Valves; EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003. Except as otherwise stated in the valve data sheets, the dimensions between the flanges shall comply with standard API 6D. However, for this purpose, the standard ASME B16.10 is applicable for all valves or sizes not listed in the standard API 6D.

A great variety of valve types are commercially available. For reasons of economy, interchangeability and minimum stock, the utilization of valve types and materials shall be in accordance with the Piping Classes.

Reduced bore shall be understood as one size below the entry port in the following diameter series: 24", 20", 18", 16", 14", 12", 10", 8", 6", 4", 3", and 2".

All valves installed on piggable lines shall have full bore opening.

Valves 2" or more in diameter shall be flanged unless otherwise specified on the data sheet.

4.6. Specialty Piping Items (SP Items)

Any piping fitting or valves not covered by the Piping and Valve Material Specification such as valve interlocks, couplings, flexible hoses, strainers, special bends, insulating flange sets, etc. if applicable in this project shall be designated as an SP item, assigned an SP number and identified on the P&IDs.

Datasheet for SP items shall be prepared to include a material description and any drawings or sketches required, for each special piping item.

SP items shall be identified and tagged with a permanently attached corrosion resistant metal tag.

The list of SP items shall be included in the overall piping MTO.

4.7. Painting and Coatings

All carbon steel pipe-work shall be painted in accordance with Specification for Painting Above-ground Facilities and specification for Coating - Buried piping.

5. Piping Drawings

5.1. Piping General Arrangements

Piping general arrangements shall be comprehensive drawings based on the piping studies, showing all piping necessary to ensure the proper function of all current and future process units of the project, and conforming to the correct flow sequence in accordance with the P&IDs. The drawings shall also show all lay-down areas, access and escape routes.

All piping shall be located by dimension, elevations and/or coordinates to the minimum sufficient to produce piping isometrics, pipe supports, and indicate influence on equipment and steelwork.

5.2. Isometrics

Isometrics shall be prepared for all lines. These drawings shall be fully detailed and dimensioned, and contain a material take-off.

5.3. Format

The following standardised formats shall be used in the preparation of the various documents:

- A0(1189 x 841 mm)
- A1(841 x 594 mm)
- A2(594 x 420mm)
- A3(420 x 297 mm)
- A4(297 x 210 mm)

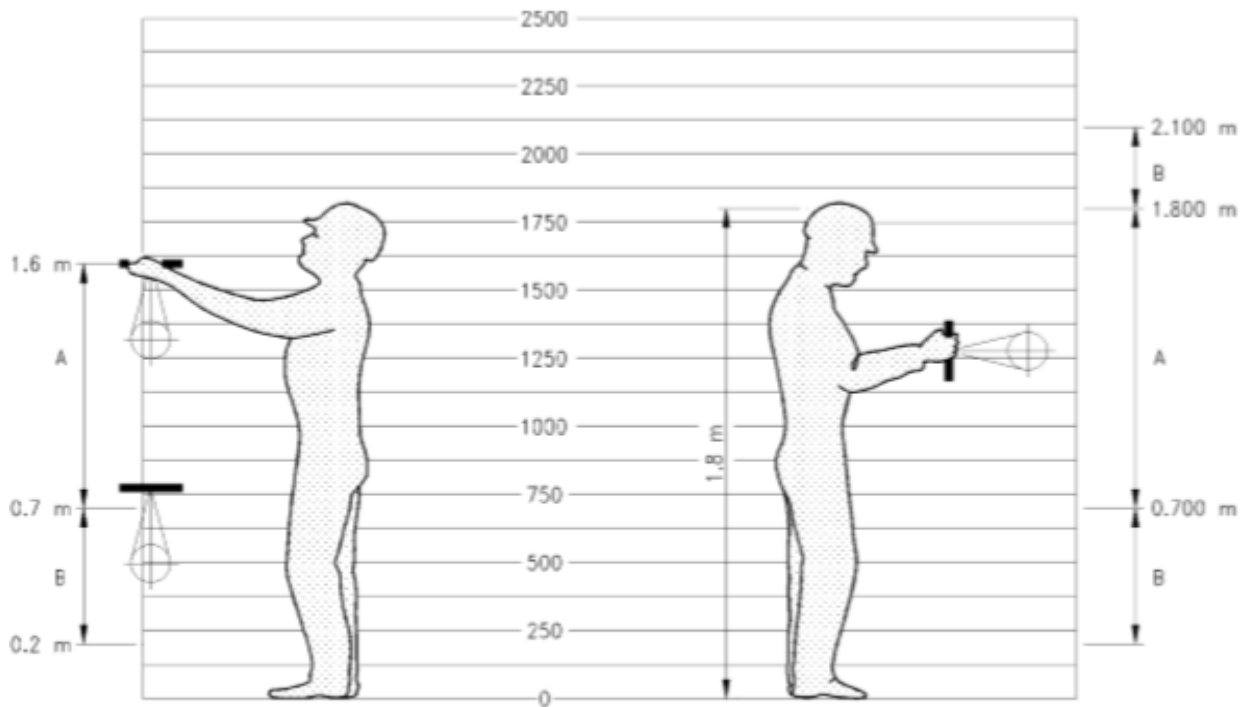
5.4. Scale

- For overall layout drawings:1/1000 – 1/500
- For unit layout drawings:1/100 – 1/200 – 1/250
- For piping and instrument diagram (PIDs):none
- For drawings of off-site piping:1/100 – 1/200 – 1/500 and 1/50 for details
- For drawings on on-site piping:1/33 1/3 – 1/50
- For isometric drawings of piping:none

The metric system shall be used for all drawings. However, the diameter of pipes shall be stated in inches (")

5.5. Appendix A: Ergonomic Valve Positioning

(DEP 31.38.01.11-Gen)



Note: Dimensions are based on personnel average height of 1.8m (± 4 cm)

5.6. Appendix B: Pipe Support Spacing

(DEP 31.38.01.11-Gen.)

CARBON STEEL AND HEAVY WALL STAINLESS STEEL

The data below are applicable to:

- Carbon steel pipes, STD wall and heavier, with a maximum temperature of 350 °C.
- Austenitic stainless steel pipes, schedule 40S and heavier, DN < 400, with a maximum temperature of 350 °C.
- Duplex stainless steel pipes, schedule 10S and heavier, with a maximum temperature of 280 °C.

DN	Maximum Span (mm) (NOTES 1 and 2)			
	Vapour service		Liquid service	
	Bare	Insulated (NOTE 3)	Bare	Insulated (NOTE 3)
25	3 850	2 300	3 450	2 250
40	4 750	3 000	4 100	2 800
50	5 350	3 600	4 550	3 300
80	6 550	4 600	5 450	4 200
100	7 500	5 550	6 100	4 900
150	9 150	6 800	7 100	5 800
200	10 500	8 050	7 950	6 700
250	11 800	9 050	8 700	7 400
300	12 900	9 800	9 150	7 800
350	15 150 (NOTE 4)	11 850	10 850	9 300
400	16 250 (NOTE 4)	12 850	11 200	9 750
450	17 250 (NOTE 4)	13 750	11 500	10 150
500	18 200 (NOTE 4)	14 450	11 750	10 400
600	18 950 (NOTE 4)	16 050	12 150	10 950

- NOTES:
1. Spans are based on straight pipe, other configurations shall be multiplied by a shape factor (see sketch below).
 2. Free draining pipes with a slope less than 1.5 mm/m require an additional check of the span.
 3. The weight of insulation and sheeting is based on insulation thickness varying from 70 mm for DN 25 to 200 mm for DN 600 and a density of 190 kg/m³.
 4. Spans limited by deflection. All other spans are limited by longitudinal bending stress.

